JCP campaign notebook

Continuous coordinate-network bases for Petrov–Galerkin reduced models

Tunable-NM-ROM project

Rebuilt October 22, 2026

What this is. The working notebook for the JCP paper: one entry per update, newest last, built by `notebook/build.sh`. It records decisions, designs, results (generated, never typed), and anything retracted. The canonical chronology stays in `LAB-LOG.md`; the plan is `reports/2026-10-06-jcp-offmesh-paper-plan.md`; the outline is `reports/2026-10-06-jcp-combined-paper-outline/`.

Contents

1 2026-10-06 — Campaign start and Phase 0 3

2 2026-10-08 — Launch: accuracy lanes (C) and two mechanism checks (A) 6

3 2026-10-08 — Mechanism: the gain is the exact gradient (A1, A2) 7

4 2026-10-08 — Second-order time stepping (C2, complete) 9

5 2026-10-09 — Wider bank (C1, complete): memory wall gone, accuracy gain not yet 11

6 2026-10-09 — Derivative-aware bank (C3, complete): the bank is not the bottleneck 13

7 2026-10-14 — Second-order references (E0, complete): 3D pass, 2D pass with flags 15

8 2026-10-15 — Step 2: second-order stepping with the wide bank (complete) 17

9 2026-10-18 — 3D speed lane (complete): speed-up over our own baselines, but the spectral FOM wins (K2 fires) 19

10 2026-10-18 — Priority 1: cost and accuracy against bank width r and mesh N (2D, plus 3D kernels) 21

11 2026-10-18 — Priority 2: qualification, 2D accuracy–cost fronts, observed time order 23

12 2026-10-18 — Priority 2: the 2D full-solver ladder and the crossover — K2 fires in 2D 24

13 2026-10-18 — Priority 3: head latent size K — the other reading of "k" 25

14 2026-10-19 — Priority 2: POD hyper-reduced baselines (C3) — H-base fails 26

15 2026-10-19 — Priority 3: error budget, longer horizons, sampling adequacy 28

16 2026-10-20 — Priority 3: test space, quadrature, rule families and a qualified 3D speed-up 29

17 2026-10-20 — Priority 4: new regimes — where the frozen model stops working 30

18 2026-10-20 — Priority 4: robustness — random rules, retraining seeds, hardware 31

19 2026-10-20 — Burgers 2D/3D campaign: final compilation 32

20 2026-10-22 — Improvement campaign, wave 1 33

1 2026-10-06 — Campaign start and Phase 0

Thesis

A reduced trial basis that is a *continuous, importance-ordered* function of position, $u(x) = G(x)c$, gives two properties. **Ordered:** one trained model offers an accuracy/cost dial at deployment (width R', head on or off). **Continuous:** the nonlinear term is an integral, so it can be hyper-reduced by classical quadrature (tensor Gauss, rank-1 lattices) at points that ignore the mesh, instead of quadrature fitted to snapshots on the mesh. The paper combines the discarded ICLR material (the dial) with Hari's off-mesh quadrature study. The ICLR paper will not be submitted.

Decisions (user, 2026-10-06)

topic	decision
Paper	One combined JCP paper; ICLR dropped.
Off-mesh code	Copied byte-identically into each lane's `vendor/` with `PROVENANCE.json` (not forked from the quadrature branches).
Full-order comparators	Run *and compare against* all three of coarse-grid, second-order and spectral solvers, in addition to Newton–BiCGStab/GMRES and IMEX-CG. Supersedes the 2026-09-21 "named iterative only" rule.
Examples	Keep the current PDEs and add harder ones (chosen in the lane designs).
NS 3D	Keep the lane and keep improving its coordinate-network bank.
Worktrees	Claude names and creates them without asking (now in `CLAUDE.md` and `AGENTS.md`).
Notebook	This LaTeX notebook, rebuilt after every update.

Phase 0 checklist

- [done] Plan and five planning notes, revised after a Codex audit: `reports/2026-10-06-jcp-offmesh-paper-plan.md`.
- [done] Paper outline (PDF): `reports/2026-10-06-jcp-combined-paper-outline/`.
- [done] Literature check of the unread preprints and a 2024–2026 sweep (plan appendix 07); see below.
- [done] Week-1 lanes created (table below).
- [done] `CLAUDE.md`, `AGENTS.md` and memory updated with the decisions.
- [user] Withdraw the ICLR submission on OpenReview if it is still active.
- [user] Agree authorship and the split of work with Hari; get his consent to use his code and results.

Literature check: outcome

None of arXiv 2505.14595 (Φ-ROM, a grid-based contrast), 2507.07830 (meshless SPH projection, no hyper-reduction), 2508.21279 (energy-conservative EQP, a fitted method) or the LiCROM cubature (random mesh vertices, equal weights) pre-empts the claims. No ROM paper found uses weighted classical quadrature for the reduced residual. Per contribution:

- ordered tunable bank: **novel for ROMs**; credit nested dropout (Rippel et al. 2014), slimmable networks, POD truncation, LiCROM;
- classical quadrature replacing fitted hyper-reduction: **novel as a combination**; ingredients partly pre-empted (Monte Carlo/uniform points in Simplicits, SNF-ROM, Weder et al.; lattices in PINN training; fitted off-mesh points in CECM);
- decoupling from the mesh's $O(h)$ error: **partly pre-empted** (SNF-ROM has the mechanism); Ingimarson–Rebholz–Iliescu (CMAME 2022) argue the opposite position and must be addressed;
- rule-family behaviour on tensor sine tests: **novel in the ROM setting**, as known theory with a new consequence;
- data-free error indicator: **partly pre-empted in principle**; call it an indicator, never a certificate.

Still to read before submission: arXiv 2607.27471, 2607.10965, 2602.01397, 2504.15657.

Week-1 lanes

All fork from `exp/2026-10-01-ns3d-coordnet-bank` @ `21c175a1b`; each mirrors to `origin/codeonly/exp/<lane>` with its own `sync_github.sh`, and vendors 69 files of off-mesh code, each checked blob by blob against its source commit (`exp/2026-10-01-quadrature-study` @ `84d28e303`, `exp/2026-10-01-quadrature-burgers3d` @ `aab6b6927`).

lane (`worktrees/`, branch `exp/...`)	experiments	role
`2026-10-06-jcp-references`	E0	order-verified second-order references; pseudo-spectral cross-check; sealed and out-of-distribution cohorts in a read-only store
`2026-10-06-jcp-mechanism`	E1, E2	rule ladders against theory; second-order stencil tensor, mesh-node analytic gradient, coarse-trained bank, all-continuum arm
`2026-10-06-jcp-fom-pareto`	E4	full-order cost ladders (iterative, IMEX-CG, second-order, coarse-grid, spectral); timing harness
`2026-10-06-jcp-hr-baselines`	E5	POD-LSPG/Galerkin with ECSW, Q-DEIM, GNAT; POD + exact tensor; best-practice EQ; CROM-style sampler

Harder examples: candidates for the designs

- time-dependent reaction–diffusion with e^u (Frank-Kamenetskii): non-polynomial, so no tensor exists;
- porous-medium diffusion $\nabla \cdot (u^m \nabla u)$ with non-integer m;
- Burgers at lower viscosity / narrower bumps, to find where the rule size and bank floor break;
- a non-box domain (e.g. L-shape with a distance-function Dirichlet factor and a composite rule);
- NS 3D with an improved coordinate-network bank.

Carried-forward corrections (from the Codex audit of the plan)

Retracted: the explicit Gauss strip-width bound in planning note 03; "lattices converge like m^{-3}" as an asymptotic rate (only a conditional upper bound); Hari's "super-algebraic lattice" and "tent adds a kink" statements for this integrand; the per-lane sealing of test cohorts (now global); the 5 % timing fault under a 10 % gate; Gauss "exactness" on sines; the rule-doubling "certificate"; the 5–7-day schedule (at least 7.6 days on four GPUs).

Next (Phase 1)

For each week-1 lane: write `DESIGN.md` → Codex design audit → code → Codex code audit → real-size smoke with every control firing → cluster jobs → Codex results audit → generated report and a notebook entry. Lanes run as separate subagents, one worktree each, with distinct cluster namespaces.

2 2026-10-08 — Launch: accuracy lanes (C) and two mechanism checks (A)

Decision (user)

Push accuracy first: run the three Group C directions, and from Group A only the two most important checks. Every step in every lane is gated by a Codex adversarial check (user instruction); the lane rules are in `reports/2026-10-08-jcp-lane-agent-rules.md`.

Lanes running

lane	question	cluster cap
`2026-10-08-jcp-wide-bank` (C1)	Now that off-mesh memory is linear in R', how far does a wider bank push accuracy? 2D at $R' = 512$; trimming the test count M; training a 3D bank with $R = 1024$; the error/cost scaling law.	1 job
`2026-10-08-jcp-time2` (C2)	Does second-order time stepping (Crank–Nicolson, BDF2) remove the backward-Euler error and allow larger steps? Includes a fair second-order full solver.	1 job
`2026-10-08-jcp-smooth-bank` (C3)	Does a bank trained for its derivatives (Sobolev loss, smoother features) lower the error floor and need fewer quadrature points? Includes a coarse-data control.	1 job
`2026-10-06-jcp-mechanism` (A1, A2)	Is the gain from leaving the mesh or from the exact gradient (mesh nodes with the analytic gradient)? Does the old error follow the stencil order (upwind vs central gap slope)? A3/A4 deferred.	1 job

Corrections recorded

Retracted: "Our trained bank already has more columns than we used" was wrong for 3D: the 3D bank has exactly 512 columns, all used. A wider 3D bank must be trained; only 2D has headroom (384 of 512 used).

Status

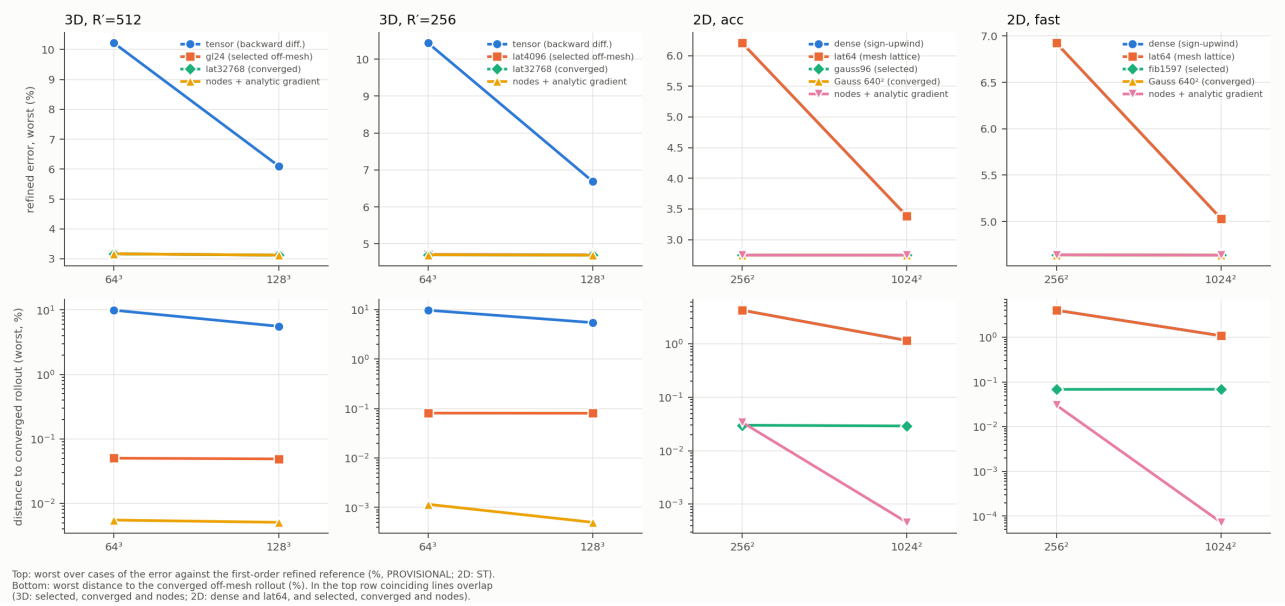
All accuracy numbers from these lanes are *provisional* until the second-order references (E0) exist. The Group B lanes (references, full-solver cost ladder, POD baselines) are not started.

3 2026-10-08 — Mechanism: the gain is the exact gradient (A1, A2)

Lane `2026-10-06-jcp-mechanism` (last commit `712d682d8`, mirror `977b28357`); jobs 5012942 (A1 3D), 5015576 (A1 2D), 5016189 (A2), all on H200, f64, `jax_backend=gpu`. Every step passed a Codex gate; the final report audit had no WRONG items. Report: `worktrees/2026-10-06-jcp-mechanism/experiments/jcp-mechanism/report.md`. **Preliminary; accuracy against first-order references is provisional.**

A1: mesh nodes with the exact gradient

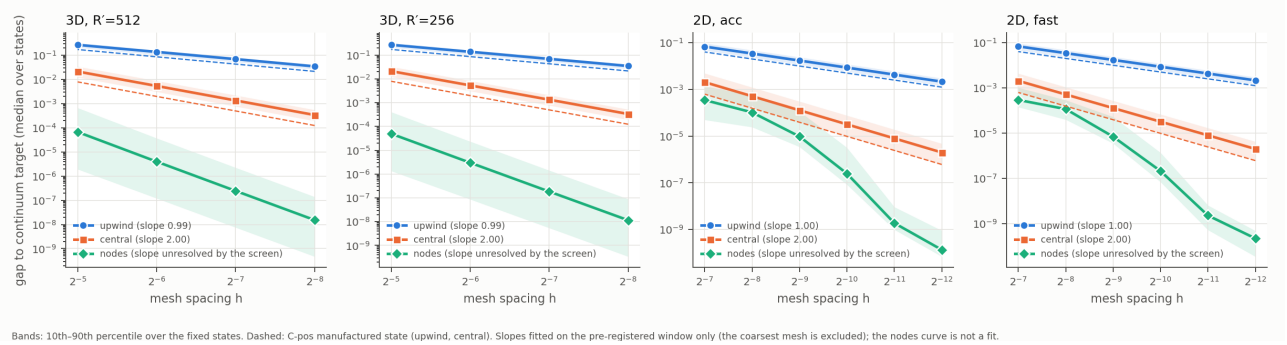
A control arm keeps every mesh node but uses the bank's analytic derivative instead of the upwind stencil. In 3D (64^3, 128^3; $R' = 512, 256$; 64 validation cases) it reproduces the converged off-mesh rollout in all four cells (pre-registered label R): the old tensor's worst error is about 10 % (64^3) and 6 % (128^3), while the mesh-node arm, Gauss 24^3 and the 32768-point lattice all sit near 3.1 %. In 2D the pre-registered rule could not be applied (label X0: the dense-vs-converged gap was below the 1 % threshold), but descriptively the mesh-node arm is again far closer to the converged rollout than the dense stencil solve.



Top: worst over cases of the error against the first-order refined reference (%, PROVISIONAL; 2D: ST).
Bottom: worst distance to the converged off-mesh rollout (%). In the top row coinciding lines overlap (3D: selected, converged and nodes; 2D: dense and lat64, and selected, converged and nodes).

A2: the stencil gap follows the stencil order

On fixed saved states, the gap between the mesh sum and the continuum integral falls with slope 0.99–1.00 for the first-order upwind stencil and 2.00 for second-order central differences, inside the pre-registered bars in all four panels, with both controls behaving. With the exact gradient the gap reaches the reference's noise floor, so its slope is not resolved.



What this changes in the paper

Kill criterion K1 fired in its useful form. The accuracy gain comes from evaluating the continuum term with the exact gradient, not from placing points off the mesh.

- The claim becomes: exact-gradient *continuum evaluation* gives the accuracy.
- Classical quadrature (thousands of points, independent of the mesh) gives it cheaply. The mesh-node arm needs every node: about 2.5×10^5 at 64^3 and 2×10^6 at 128^3.

Changed or corrected along the way

- The 2D finite-difference gradient gate was redesigned as a convergence check (the bank is steep near the walls; the analytic gradient is correct).
- Historical cohort hashes are now report-only, because they include GPU floats.
- A non-finite target now invalidates a result rather than being retried.
- A halved A2 state set was caught by the code audit before the job ran.

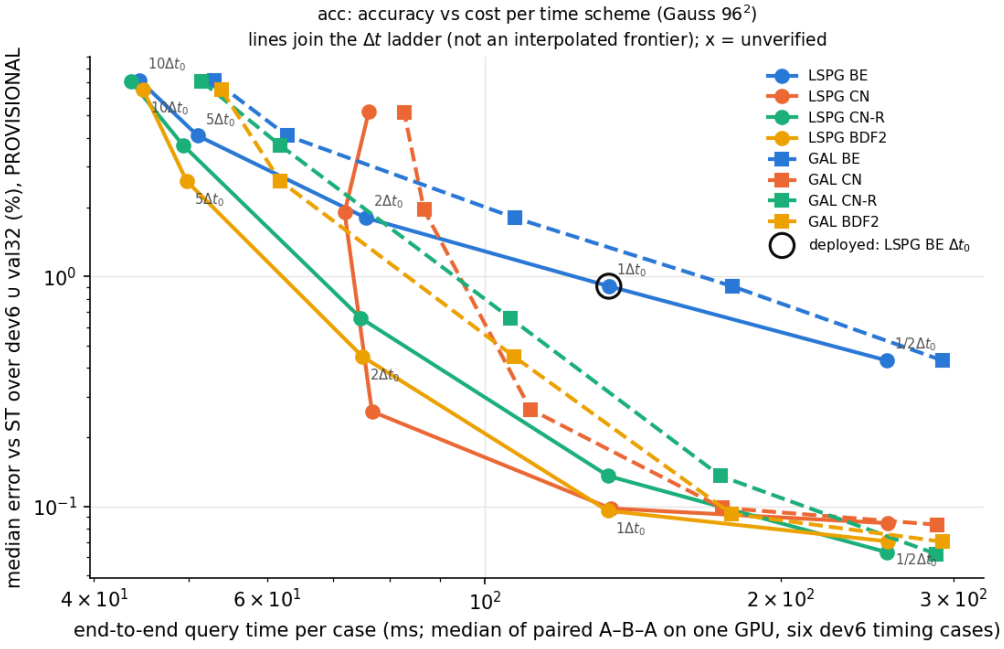
Open: whether to pre-register a new 2D separation rule, and a solver-sensitivity rerun at 128^3. A3 and A4 remain deferred.

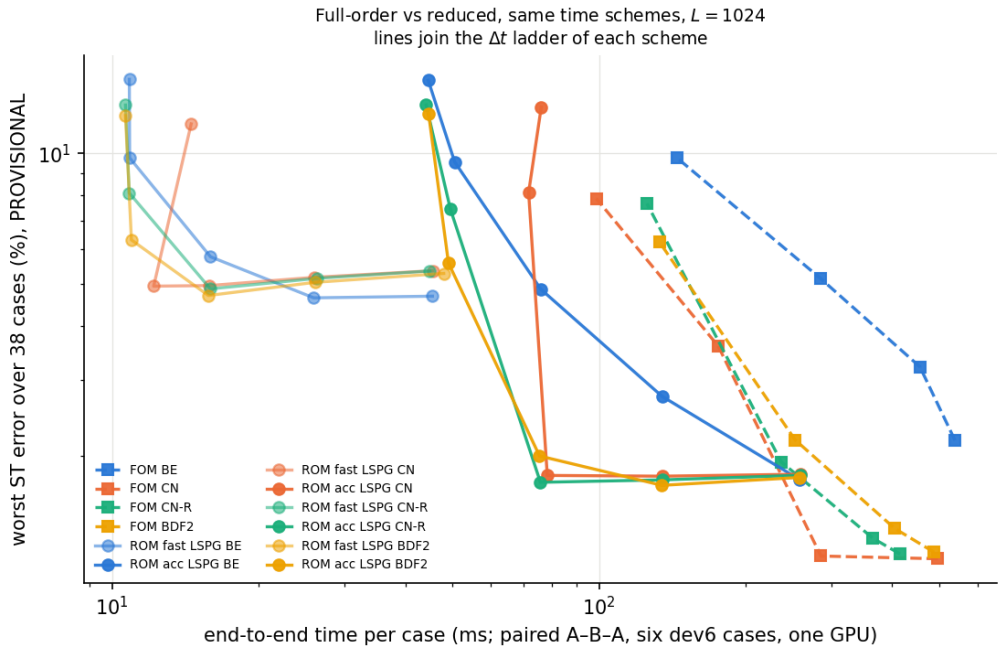
4 2026-10-08 — Second-order time stepping (C2, complete)

Lane `2026-10-08-jcp-time2` (last commit `a7a12528f`, mirror `1445e4735`). Jobs 5012794, 5012974, 5015584, 5017631 (A100), 5020049, 5023437 (H200); all audited, namespace emptied. Every step was Codex-gated, and the final report audit came back clean. Report: `worktrees/2026-10-08-jcp-time2/experiments/jcp-time2/report.md`. **All accuracy numbers are provisional** (first-order references).

2D, accurate setting ($R' = 384$, Gauss 96^2, 1024^2, 38 cases)

- Same step and cost: worst/median error vs ST goes from 2.747/0.906 % (backward Euler) to 1.796/0.098 % (Crank–Nicolson). Time error falls from 0.978 % to 0.026 %.
- At twice the step, Crank–Nicolson gives 1.803/0.259 % and runs about 1.75× faster. Both pre-registered hypotheses pass.
- Fair comparison: the full solver given the same schemes is also faster and more accurate with Crank–Nicolson. Matched speed-up against it is 1.76× (backward Euler) and 3.65× (Crank–Nicolson at $2\Delta t$).





Other settings

- Wide ($R' = 512$): 2.837/0.907 % → 2.166/0.111 %; both hypotheses pass.
- Fast ($R' = 128$): the median improves (1.035 → 0.419 %) but the worst case worsens (4.637 → 5.171 %); both fail, because the small bank's error dominates.
- 3D (65^3, 16 cases): time error drops 21× (1.87 → 0.089 %) at about equal cost. The generic solver is about 1.7× the deployed fixed sweep, so the gain is accuracy, not speed. The 3D reference is itself backward Euler at $\Delta t/4$, so it cannot rank the schemes (pre-registered).

Retracted or corrected

Retracted: the "1–2 percentage points" of backward-Euler time error is the maximum gap; the median gap is 0.5–0.8 pp. Least-squares Crank–Nicolson is not second order and must not be called so; the Galerkin form and the CN-R/BDF2 least-squares variants are. Comparing against a backward-Euler full solver overstates the speed-up.

Open

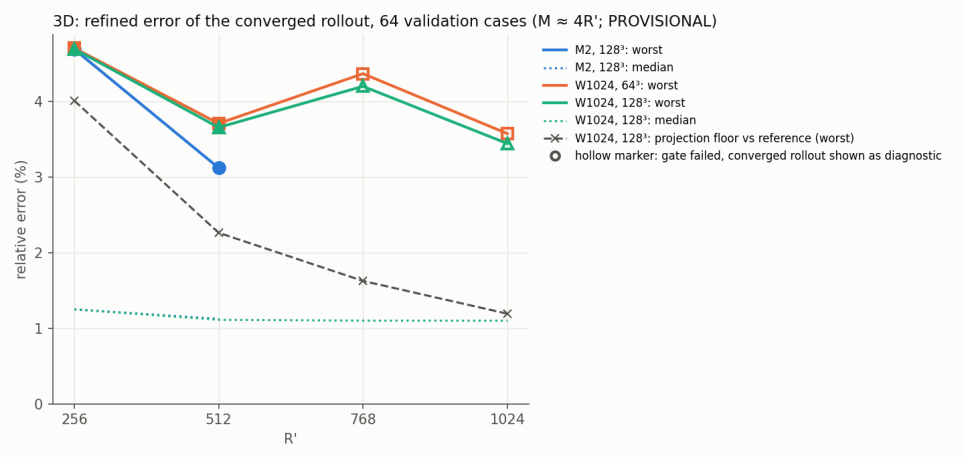
A fixed-sweep second-order solver for 3D (the route to a 3D speed-up); rerun against second-order references; choose the Galerkin or least-squares form for the paper; test64 and 4096^2 replication.

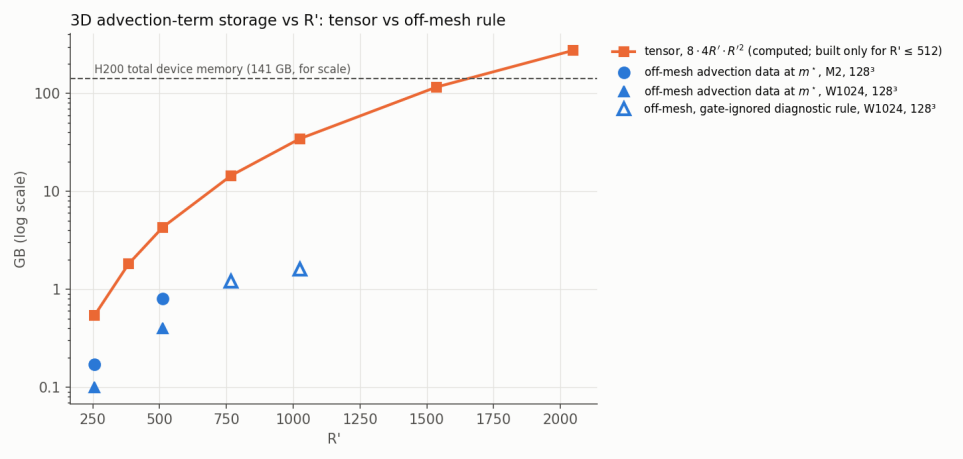
5 2026-10-09 — Wider bank (C1, complete): memory wall gone, accuracy gain not yet

Lane `2026-10-08-jcp-wide-bank` (last commit `2c3c397e0`, mirror `dbb3da96f`). Jobs 5012313, 5012763 (A100); 5012847, 5022387 (H200); 5020724, 5020855, 5021867, 5022113 (H100). Every step was Codex-gated, and the final report audit says REPORT-OK. Report: `worktrees/2026-10-08-jcp-wide-bank/experiments/jcp-wide-bank/report.md`. **Provisional:** first-order references, and the 3D selections are post hoc (A7). The J4 solver uses 6 adaptive steps (A8), so J4 is not comparable with earlier 3D runs.

Findings

- **2D dial stops at $R' = 384$.** Against the space-only reference, the worst error falls 3.45 → 1.49 → 1.05 % for $R' = 128 \to 256 \to 384$, but $R' = 512$ is worse on 28 of 38 cases at 3.4× the cost. Against the space+time reference the median is flat at 0.91 %, consistent with time-step error dominating (a hypothesis).
- **Test-count trim does not work.** The registered mechanism is contradicted in 2D and 3D; only one 2D trim gives a 25 % saving for a negligible ST change.
- **Wider 3D bank ($R = 1024$) trained.** Its worst projection floor falls 3.23 → 1.93 → 1.29 % for $R' = 512 \to 768 \to 1024$.
- **But the 3D ROM barely follows.** From $R' = 512$ to 1024 every case improves slightly, yet the median stays near 1.1 % and the worst error is non-monotone (about 3.4–4.4 %). The registered verdict is *unavailable*, because a quadrature-convergence gate fails at $R' \geq 768$.
- **The memory wall is gone.** At $R' = 1024$ the off-mesh rule needs 1.61 GB, against 34.37 GB for the tensor.





Retracted

Retracted: the 4-case preview of 1.43 % worst error at $R' = 1024$, 128^3 (relayed on 2026-10-08) did not hold on all 64 cases: the full-cohort worst is about 3.4 %. "The 2D dial extends to 512" and the registered test-count-trim mechanism are both retracted.

What it means

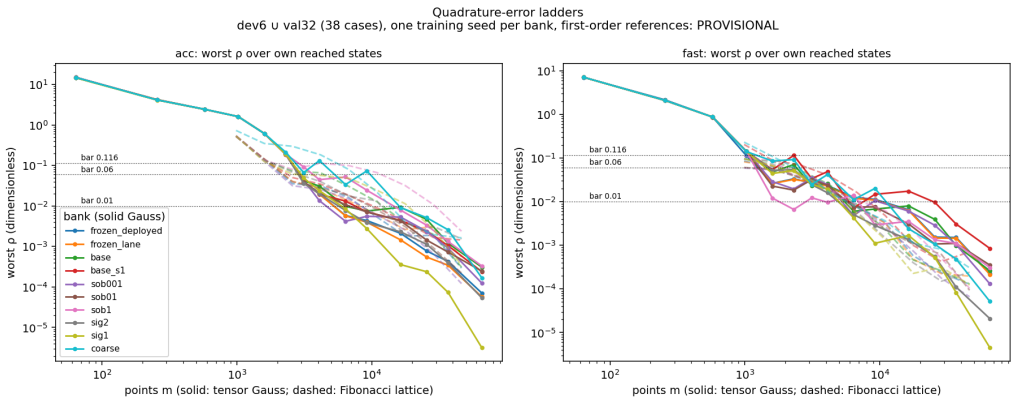
The bank is no longer the bottleneck: the reduced model sits well above the bank's projection floor. With C2 (time error dominates) and C3 (smoother or Sobolev banks do not help the ROM), the evidence points to time discretisation and the first-order reference as the next limits. The obvious next experiment is second-order stepping with the wide bank, against second-order references.

6 2026-10-09 — Derivative-aware bank (C3, complete): the bank is not the bottleneck

Lane `2026-10-08-jcp-smooth-bank`, last commit `12f620782`, mirror `8ede69cba`. Jobs 5016275 and 5027733 (training, 4× A100 each); 5027022 and 5037650 (evaluation). All steps were Codex-gated. The report audit confirmed all fixes except four minor presentation items. Report: `worktrees/2026-10-08-jcp-smooth-bank/experiments/jcp-smooth-bank/report.md`. **Provisional:** first-order references, one training seed per recipe, development and validation cohorts only.

Findings

- **Smoother banks need no fewer points at the acceptance bars.** In the accurate setting every bank needs Gauss 56^2. The point count there is set by the $M = 1536$ high-frequency tests, not by the bank: with only the first 320 tests, Gauss 32^2 is already near the bar.
- **Smoothness pays only in the high-accuracy tail.** At 256^2 points the worst ρ is 3.2×10^{-6} for $\sigma = 1$, against 3.1×10^{-4} for the baseline, with no loss of representation accuracy.
- **Sobolev training improves the bank's derivatives but not the ROM.** The median gradient error falls by 53, 70 and 74 % for $\lambda = 0.01, 0.1, 1$. The ROM error against the space-only reference does not improve (it rises by 62 % at $\lambda = 1$).
- **All banks sit at the time-stepping floor** against the space+time reference: 2.74–2.97 % in the accurate setting.
- **Coarse-data control.** A bank trained only on 129-node data gives 0.83 % worst error against the space-only reference, against 7.58 % for the 129-node FOM and 4.12 % for the 257-node FOM, in the accurate setting. So finer training data were not necessary for the gain. This holds in the accurate setting only, and the cause is not isolated. The same bank needs 16384 points.



Corrections

Retracted: runs/SUMMARY-R3.md describes the wrong training job; the deployed bank is r3a. The 2D lane's training-overlap check used the wrong draw (the cohorts were re-verified as disjoint). Retrains are not bitwise reproducible without deterministic XLA flags. In the fast setting, the 2D numbers depend on the column-ordering procedure (3.45 % vs 4.51 %).

Week summary (8–9 October)

- **Mechanism (A):** the accuracy comes from the exact gradient; classical quadrature makes it cheap.
- **Second-order time stepping (C2):** the clear win. The median error falls about 9× at equal cost, and the matched speed-up against a fair second-order FOM rises from 1.8× to 3.7×.
- **Wider bank (C1)** and **smoother or Sobolev bank (C3):** each represents the solution better, but the ROM error does not follow. The bank is no longer the bottleneck.
- **Next:** second-order stepping with the wide bank, measured against second-order references (the E0 lane); a fixed-sweep second-order 3D solver; a test space with lower mixed frequencies.

7 2026-10-14 — Second-order references (E0, complete): 3D pass, 2D pass with flags

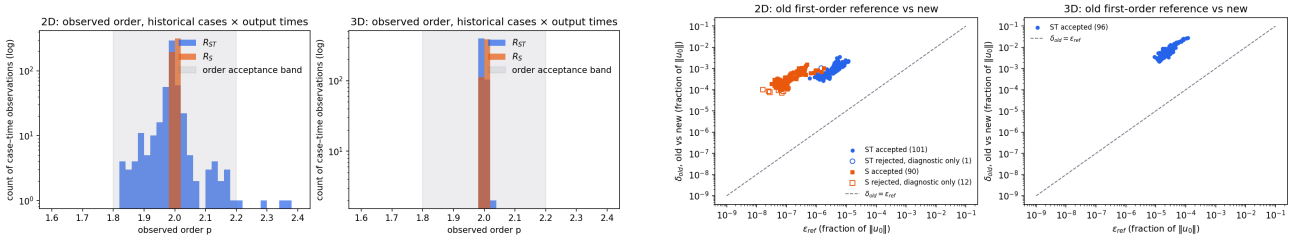
Lane `2026-10-06-jcp-references`, last commit `cc04f4b54`, mirror `66ee55015`. Cluster namespace `jcp-ref`; 39.7 GPU-h (A100 and H200). Every step was Codex-gated, and the final report audit is READY. Report: `worktrees/2026-10-06-jcp-references/experiments/jcp-references/report.md`. Loader: `refstore.py`. Numbers below are quoted from the lane's generated report.

Findings

- **3D passes.** All 200 historical case-kinds pass the order check, with observed order 1.99–2.02. The median ϵ_{ref} is $\approx 2.5 \times 10^{-5}$ and the worst is 1.16×10^{-4}. The pseudo-spectral cross-check passes 8/8, with the measured distance equal to 1.00–1.02 of ϵ_{ref}. Sealed cohort 128/128, OOD 32/32.
- **2D passes with flags.** 191/204 historical case-kinds pass, with $\epsilon_{ref} \approx 3$–4×10^{-6} (space+time) and $\approx 1.2 \times 10^{-7}$ (space only). The 13 flagged cases are not rankable:
  - 12 high-viscosity space-only cases fail the (conservative) solver-floor check;
  - test64 case 36 fails the order check.

  The cross-check passes 8/8. Sealed cohort 230/256 accepted, OOD 64/64.
- **How wrong the old first-order references were** (maximum over output times):
  - 3D: median 0.7 %, maximum 2.81 %;
  - 2D: median 0.03–0.08 %, maximum 0.35 %.

  So 3D numbers can move noticeably on re-scoring, and 2D numbers barely.
- **Controls fired.** A deliberately first-order solver and an injected first-order defect are both rejected. The cross-check against the old references fails on 16/16 cases.



Caveats and corrections

Retracted: Amendment A10 replaced the pre-registered cross-check rule (pseudo-spectral order ≥ 2.5) after the pseudo-spectral data had been seen, though before any comparison with the new references. Under the original rule, only 1/8 2D cross-checks would pass, and 2D would be E0 FAIL. Both verdicts are reported per case. The earlier "1.7–2.3 %" for the old 3D reference error was too low: the maximum is 2.81 %. ϵ_{ref} is a conditional Richardson estimate, not a bound. Sealing is cooperative only: the key sits on the shared cluster account. The sealed cohorts are generated but **not opened**, and no ROM has been scored on them.

Open

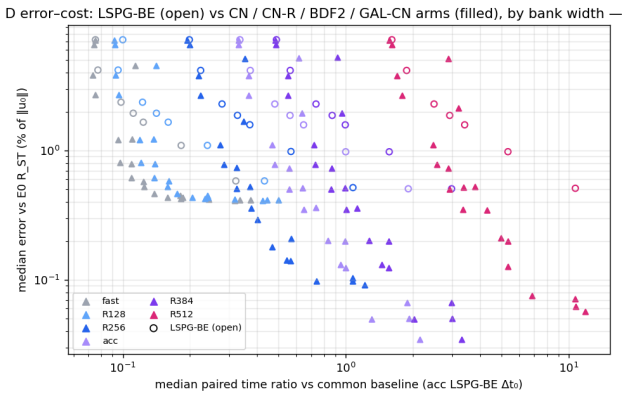
- Key custody.
- Whether to revisit the 2D floor check before opening the sealed cohort.
- Building the opening tool.
- A CN-matched space-only reference.
- Re-scoring every earlier provisional number against this store.

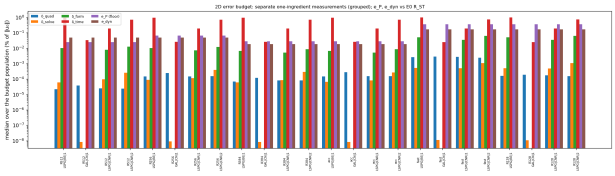
8 2026-10-15 — Step 2: second-order stepping with the wide bank (complete)

Lane `2026-10-13-jcp-combined`, last commit `a3c9e170d`, mirror `c51cecdbd`. Namespace `jcp-comb`, now empty; the jobs are in `runs/JOB-LEDGER.json`. There are 44 Codex audits, and the final report audit is READY. Report: `worktrees/2026-10-13-jcp-combined/experiments/jcp-combined/report.md`. Primary scoring is against the E0 second-order references, with E0's conditional label. First-order scores are kept beside them and stay provisional. The 3D rule selection is **post hoc** (A10, A11). Numbers are quoted from the generated report.

2D (38 development and validation cases)

- **Selected scheme:** LSPG-CN-R. At equal cost, the accurate arm's median error falls from 0.98 % (BE) to 0.20 % (`acc` setting), and the worst from 3.0 % to 1.8 %.
- **Bank width (H1):** R128→R256 passes (median −53.7 %, 38/38 cases improve). R256→R384 fails the 25 % bar (−5.2 %). R384→R512 does not pay off.
- **Tunability (H2):** passes.
- **Speed against the second-order FOM (H5):** the selected scheme's best matched speed-up is 13.6×. The overall best is 17.7×, from an LSPG-CN arm that is not second order. The re-measured C2 `acc` arms reach 3.89×. The FOM is second order in time only.
- **H3:** the error-cancellation explanation for the fast setting's worst case is *not supported*.
- **Finalist quadrature check:** the selected arms fail it at `fast`, `acc` and R256. The fallbacks for the fast arms qualify; those for the accurate arms cost 1.3–1.6× and do not.

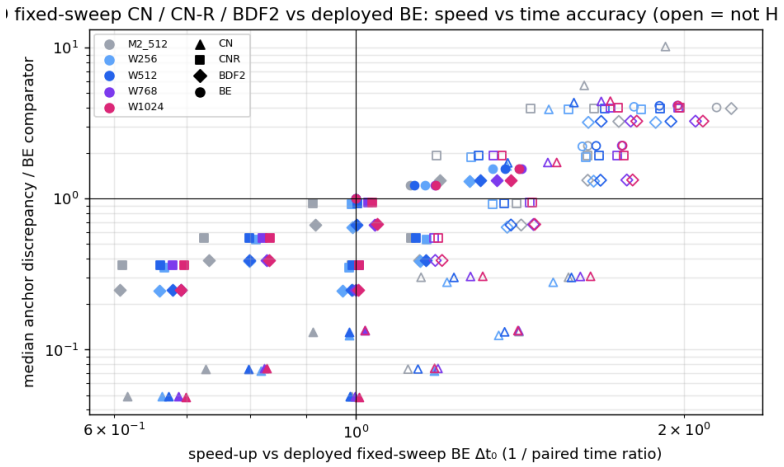




3D (validation cohort 923801, 64 cases, 65^3)

- **Accuracy at equal cost (H4b):** passes at every width. At W1024 the selected arm's median error is 0.12 %, against 1.53 % for the deployed BE. The worst errors of the selected arms are still 4.4–6.7 %.
- **Speed at equal accuracy (H4a, ratio ≤ 0.6):** fails at every width. The closest equal-accuracy arms reach only 0.85–0.96.
- **Width dial (H6):** W512→W768→W1024 extends (median −60 %, then −53 %). W256→W512 does not.

- **J4 gate failure at $R' \geq 768$:** caused by one case (case 5). W1024 converges only at Gauss 80^3.
- **Fixed-sweep CN (K=2, $2.5\Delta t_0$)** fails on a fixed set of 7 cases, so the pre-registered rule selection was unavailable.



What limits next

Time discretisation is still the largest budget component (2D: 16 of 24 setting-arms; 3D: 13 of 19), followed by the bank's projection floor.

Corrections

Retracted: The pre-registered 3D rule selection was unavailable; the one used is post hoc (A10, A11). A first frozen 3D fast-equal-accuracy selection was retracted before any budget job (A15). Some first-order H1 and H6 verdicts change under E0 (A13). Earlier drafts used second-order wording for the reduced arms and stacked non-additive budget bars. Offline scoring on the local GB10 GPU corrupted R'=1024 decodes, so all offline analysis now runs on CPU.

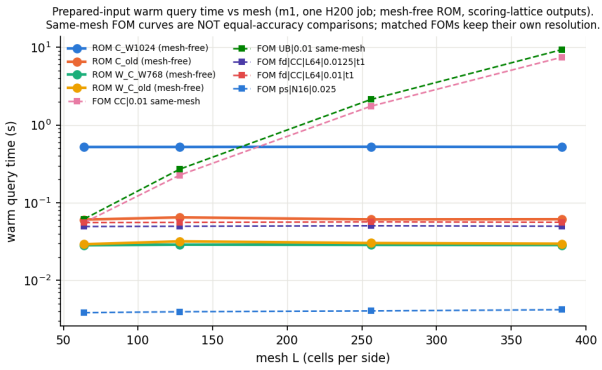
Follow-up

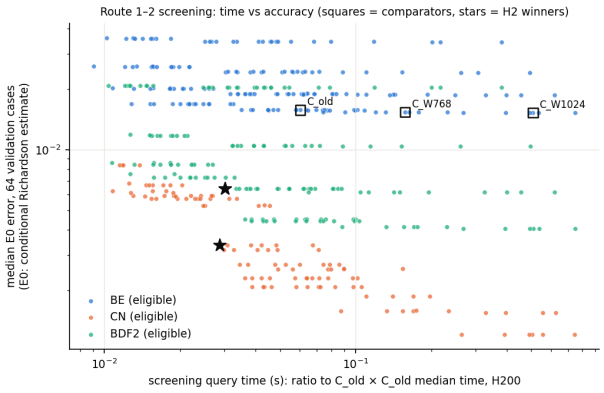
The 3D speed lane (`2026-10-14-jcp-speed3d`) is running. It tries one-pass Jacobian reuse at large steps, narrower banks, overhead cuts and batching, and large-mesh FOM comparisons, including a pseudo-spectral FOM.

9 2026-10-18 — 3D speed lane (complete): speed-up over our own baselines, but the spectral FOM wins (K2 fires)

Lane `2026-10-14-jcp-speed3d`, last commit `691465d88`, mirror `6c270e6c5`. The lane was resumed by the Burgers campaign coordinator after its first agent stopped. Jobs on H200 (all `jax_backend=gpu`): g1 5301580, g2 5301264, g3 5301270, f1 5301274, p1 5352455, m1 5352457, r1 5353792; 10.9 of 34 GPU-h. Namespace emptied. Final report audit READY. Numbers are quoted from the generated `report.md`; errors are against E0 R_{ST} (a conditional Richardson estimate).

- **Speed over our earlier 3D configurations (H2, bar 1.67×, paired panels):** pass against all three. 2.01× vs the old deployment, 5.37× vs W768, 17.2× vs W1024. The winners use the narrower M2_512 bank with Gauss 24^3 and second-order stepping (BDF2 one-pass or quasi-Newton CN).
- **Cost is flat in the mesh (H4.1):** the online solve time changes by at most a factor 1.007–1.09 from 64^3 to 384^3 for all four ROMs (mesh-free query; deployment mode policy). Coordinator recheck from raw m1 timings: 1.004–1.098, same verdict.
- **Against the full simulation: K2 fires.** At 384^3, no selected ROM reaches 2× against the best matched FOM across families; against that union the ROMs run at 0.008–0.16× the FOM's speed. A pseudo-spectral FOM at $N = 16$ matches the ROM's accuracy. Against central-CN only, one ROM reaches 2.03×. Matched speed-ups are upper bounds (finite FOM ladder).
- **The early observation** "spectral FOM clearly beats the ROM at 64^3" is **confirmed**.
- Replication on cohort 923901: all three winners hold (2.03×, 5.31×, 16.6×).





Consequence for the paper

Following the plan's written consequence for K2: report the speed-up honestly (no win over the spectral FOM in 3D Burgers on the box) and lead with mesh-independence of cost. The upwind FOM cells are undecided (no upwind ladder member is as accurate as the ROM).

Corrections

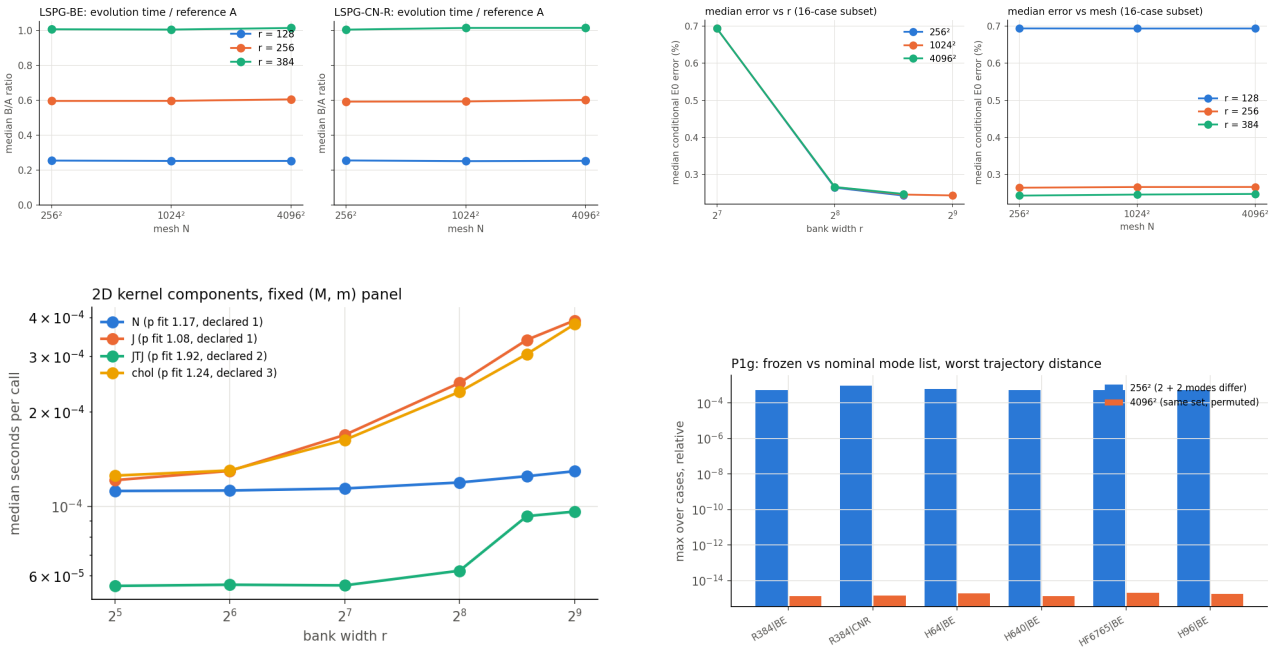
Retracted: A9's attribution of the G2 failures was wrong (corrected in A10). G2 fails literally on 15 of 288,944 steps, all at $\gamma \leq 3.4 \times 10^{-12}$; no classification changes. The m1 calibration population and the L256 resource estimate were corrected before results were read. The flat-cost result holds for the

deployment mode policy, not for a fixed test space (no fixed-mode panel ran). Replication eligibility is not a fresh Q1 certificate.

10 2026-10-18 — Priority 1: cost and accuracy against bank width r and mesh N (2D, plus 3D kernels)

Lane `2026-10-17-jcp-p1-scaling`, last commit `5b782026b`, mirror `3b589dad5`. Eleven jobs (A100/H200, all `jax_backend=gpu`), 1.91 of the 20 GPU-h cap; nothing deferred. Design restated by the coordinator from plan v4 and Codex-audited (4 rounds) before launch; lane audits: design 5 rounds, code 4, jobs 5, results/report 4, all clean at the end. Numbers quoted from the generated `report.md`. Errors are against E0 R_{ST} (conditional Richardson ε_{ref}).

- **Solve cost is mesh-independent (H-cost-N): pass, 8/8 configurations** ($r = 128, 256, 384$; BE and CN-R; 256^2, 1024^2, 4096^2). Max/min solve time over meshes 1.01–1.02, every 95 % interval ≤ 1.04 (bar 1.25). Coordinator recheck from raw timings: 1.0085–1.016.
- **Accuracy is mesh-independent (H-acc-N): pass, 8/8.** Every case within its margin (38 cases at 256^2, 16 at 4096^2). R128 CN-R shows a small detectable trend ($p = 0.0063$), so "pass" there means practical equivalence within the margin.
- **Accuracy improves with r (H-acc-r):** 128→256→384 improves at every mesh and for both schemes (majority of resolved cases). 384→512 at 1024^2 does *not* improve (BE 4 better / 16 worse / 18 tied; CN-R 6/22/10): the known regression is reproduced.
- **Kernel cost against r (H-cost-r): 7 of 8 component fits fail** their declared operation-count exponents. Cholesky grows like $r^{1.24}$ (2D) and $r^{1.03}$ (3D) instead of r^3; $J^\top J$ like $r^{1.92}$ / $r^{1.33}$ instead of r^2. At these sizes the GPU kernels are far from their asymptotic regime; this is a statement about measured cost, not about operation counts.
- **Qualification:** all 22 rollout configurations pass Q1 (solver) and Q2 (quadrature). The archived combined-lane 1024^2 results reproduce to $\leq 2.9 \times 10^{-15}$.
- **P1g, the frozen mode list:** the 2D test-mode set differs only at 256^2 (2 of 1536 modes swapped); at 4096^2 it is the same set in another order. Freezing it moves trajectories by 5–9×10^{-4} but errors by only 0.01–0.05 percentage points. Earlier cross-mesh claims compared slightly different test spaces at 256^2; the conclusion does not change.



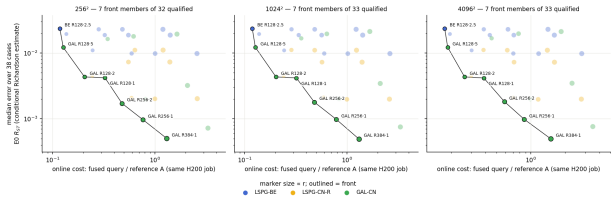
Corrections

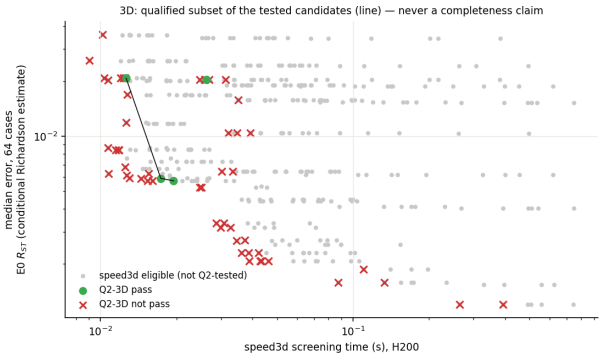
Retracted: The coordinator's restatement said the 2D mode lists also differ at 4096^2; they do not (permutation only). The plan's "1024^2 reused from combined" was false for R256 (different rule); all 1024^2 runs were redone. In the quadrature slice, `gauss64` was registered as the failing 2D rule but passes both bars (the pre-data expectation was wrong); in 3D `lat8192` fails (4.2e-3 ¿ 2.5e-4). The first smoke acceptance failed 5 checks, all defects in the checks themselves. The component-sum control was demoted to a diagnostic (deviation from AM-5, disclosed).

11 2026-10-18 — Priority 2: qualification, 2D accuracy–cost fronts, observed time order

Lane `2026-10-18-jcp-qualify`, last commit `4399491eb`, mirror `fb9b137be`. 16 jobs (A100/H200, `jax_backend=gpu`), 4.13 of 35 GPU-h, nothing deferred; final report audit READY. The frozen 2D front `runs/FRONT-2D.json` (commit `2374fafe1`) feeds the 2D full-solver comparison. Numbers quoted from the generated `report.md`; errors are E0 conditional estimates.

- **Qualification (2D):** 98 of 112 configurations pass the solver (Q1) and quadrature (Q2) checks. LSPG-CN-R at $2\Delta t_0$ fails Q1 for R128 and R256 at every mesh (worst case 3.6–4.0e-4 vs bar 2.5e-4).
- **Fronts (paired timing, one job per mesh):** the Galerkin CN scheme (registered as a diagnostic) holds 6 of 7 median-front places at every mesh; the most accurate member is R384 GAL-CN at Δt_0, 0.050 % median at 1.31× the reference cost (1.5× at 4096^2). Primary comparisons keep the registered LSPG schemes; GAL-CN is reported separately.
- **Observed time order:** LSPG-BE 1; GAL-CN 2; LSPG-CN-R 2 at R384 ($\hat{p} = 1.96$) but 1.74 at R128 (no integer label). Valid over Δt_0 to $\Delta t_0/4$.
- **Accuracy vs mesh, extended set:** 21 pass, 9 inconclusive (5 because a configuration is unqualified). **Accuracy vs r:** the 384→512 regression recurs in every cell of the 128–512 range.
- **3D quadrature (Q2):** only 4 of 55 tested speed3d configurations pass. Both speed3d H2 winners fail (successor-rule distance 8.1e-4 and 9.5e-4); coordinator coefficient-space recheck 0.94e-3 and 1.10e-3.





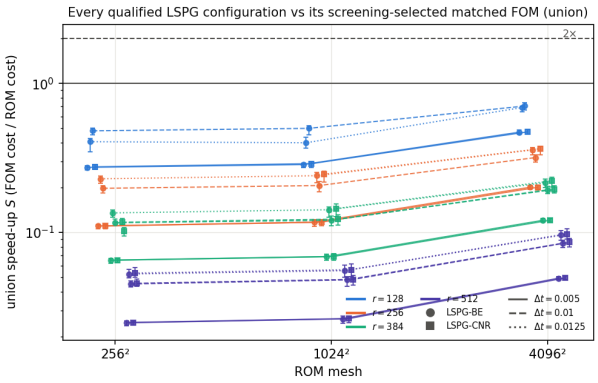
Corrections

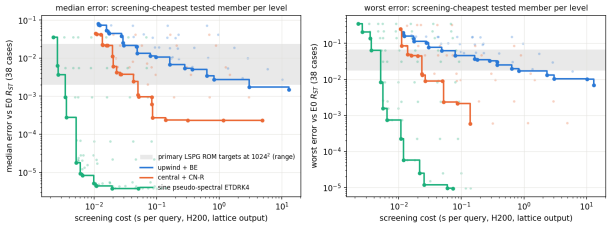
Retracted: The speed3d H2 speed-ups (2.01×, 5.37×, 17.2×) are measured correctly but their winning configurations are **not quadrature-qualified**; they must not be called qualified. The combined lane's finalist table labels the R128 and R256-fallback "fast equal accuracy" arms (LSPG-CN-R at $2\Delta t_0$) qualified; under the every-case Q1 contract they are not. For Galerkin the stationarity ratio was replaced by a residual criterion (disclosed deviation).

12 2026-10-18 — Priority 2: the 2D full-solver ladder and the crossover — K2 fires in 2D

Lane `2026-10-18-jcp-fom2d`, last commit `602b1de51`, mirror `e10358562`. Nine H200 jobs (`jax_backend=gpu`), 4.59 of 30 GPU-h; final report audit READY. Numbers quoted from the generated `report.md`. Errors are E0 conditional estimates; every speed-up is an upper bound (finite ladder, screening-selected comparator); timings are latencies on dev6.

- **Ladder (C1):** 185 FOM configurations over upwind+BE, central+CN-R and sine pseudo-spectral (ETDRK4), each over mesh ($L = 32$–4096; spectral $N = 8$–256), Δt and tolerance, on 38 cases; 170 admissible. Gates: E0's own level reproduced to 9×10^{-15}.
- **Best FOM:** for every one of the 98 ROM front targets the cheapest FOM matching both median and worst error is *spectral*, at $N = 48$ or $N = 24$.
- **K2 fires in 2D:** at 4096^2, none of the 21 qualified LSPG-BE/LSPG-CN-R configurations is faster than its matched FOM; union speed-up 0.049–0.706 (largest interval end 0.74). Example: R128 LSPG-BE at Δt_0, 0.025 s vs 0.012 s for spectral $N = 48$ (coordinator recheck from raw timings: 0.0249 vs 0.0117 s). No crossover at 256^2, 1024^2 or 4096^2.
- **Against narrower comparators:** 16 of 21 are $\geq 2\times$ faster than the selected upwind+BE FOM; on the ROM's own 4096^2 mesh the ROM is 5.8–84× faster than central+CN-R. Those are statements about those comparators only.





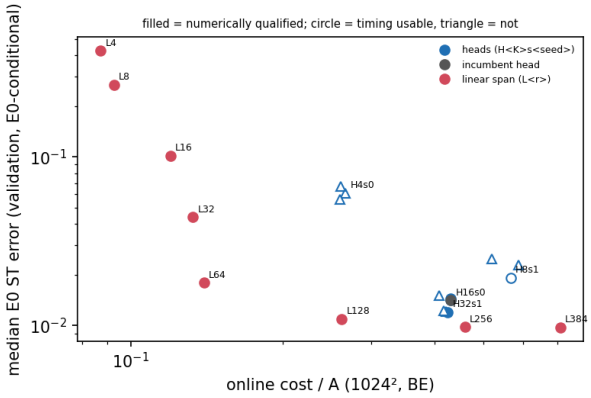
Consequence and corrections

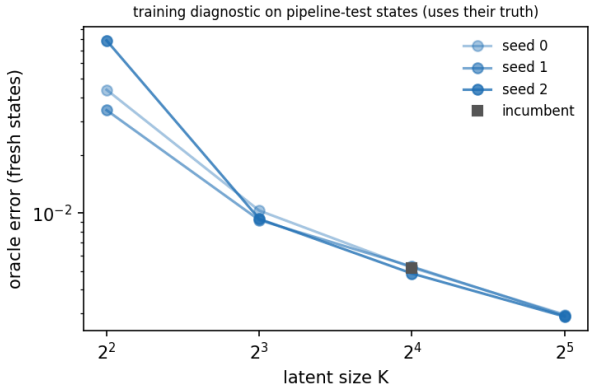
With K2 firing in both 2D and 3D, the plan's written consequence applies: no "substantially faster than the best full solver" claim; the paper leads with mesh-independent cost (and memory), and reports speed-ups against named comparators with their scope. **Retracted:** The combined lane's H5 speed-up (13.6× selected scheme, 17.7× best arm) holds only within its scope — a same-mesh upwind FOM matched on worst error — and does not survive the expanded ladder. Spectral at the ROM's own resolution and the governing plan's other solver families (upwind+CN-R, Newton–GMRES, IMEX-CG) were not run.

13 2026-10-18 — Priority 3: head latent size K — the other reading of "k"

Lane `2026-10-18-jcp-heads`, last commit `0ba9a1288`, mirror `ed3d0c841`. Nine jobs (A100/H200, `jax_backend=gpu`), 3.68 of 35 GPU-h, nothing deferred; results audit 3 RESULTS-CONFIRMED, report audit 6 READY. **Exploratory validation evidence only** (no family-wise claim); errors E0-conditional. Numbers quoted from the generated `report.md`.

- **Training:** 12 heads ($K \in \{4, 8, 16, 32\} \times$ 3 seeds), the incumbent's exact recipe and data. Oracle error falls from 3.4–7.9 % at $K = 4$ to 0.29 % at $K = 32$.
- **Qualification is the bottleneck:** at 1024^2 (BE) only $K = 16$ seeds {0, 1} and $K = 32$ seed {1} pass the every-case solver check; $K = 4$ and $K = 8$ none. So **error vs K is inconclusive**, and so is the nonlinear-manifold question (F5).
- **Head vs linear span:** at matched dimension ($K = r = 16$) the head is far more accurate (1.44 % vs 10.2 % median) but costs 3.78× more. **At no more online cost the linear span wins**: $r = 128$ reaches 1.10 % at 0.59× the head's cost. Coordinator recheck from per-case values: 1.43 %, 1.10 %, 10.17 %; the $r = 128$ value matches priority 1 (1.095 %).
- **Cost vs mesh:** flat for the $K = 16$ head (ratio 1.064) and the incumbent (1.029). Cost vs K: no identifiable exponent.





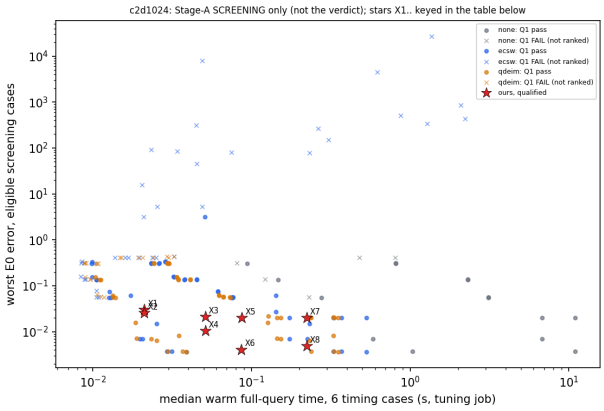
Qualified, not retracted

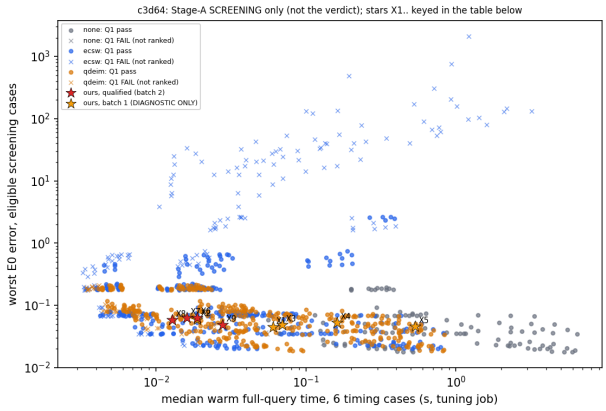
The 2026-09-14 claim "the head is necessary at matched latent dimension" still holds at matched dimension; it does not hold for accuracy per unit cost. The seed-0 $K = 16$ retrain equals the incumbent to round-off (same-seed replay, not an independent replication). Head-repair hypotheses (E3c) were not tested.

14 2026-10-19 — Priority 2: POD hyper-reduced baselines (C3) — H-base fails

Lane `2026-10-06-jcp-hr-baselines`, last commit `2e3abac47`, mirror `2fa5b5967`. Eleven H200 jobs, 21.18 of 30 GPU-h (three attempts cancelled after defects were found and fixed: a slow NNLS column deletion, a results-file rewrite storm, and a time-step parsing bug). Design 21 rounds, code 33, results 3 (CONFIRMED), report 5 (READY). Numbers quoted from the generated `report.md`. **All comparisons are developmental, in-sample and unadjusted**: both methods were tuned and compared on the same validation cases; errors are E0-conditional.

- **Baselines:** POD-LSPG with ECSW (support-targeted greedy NNLS) or Q-DEIM hyper-reduction, native and common solvers, POD snapshots from the central-difference FOM on the training draws.
- **2D 1024^2: a baseline selected by the registered role rule dominates all 8 of our qualified configurations** (faster per query and lower worst-case error, non-overlapping intervals). Like-for-like example, our R512 LSPG-CN-R: worst 2.17 % vs the ECSW pick's 1.35 % at 0.14× our time. Restriction: snapshot-limited.
- **2D 256^2:** 8 of 8 dominated (the saturation check failed; the registered all-step basis was used).
- **3D 65^3:** 7 of 9 dominated (5/5 diagnostic batch, 2/4 quadrature-qualified batch; the other 2 neutral conclusions are withheld because basis adequacy failed at ranks 768–1024). Most frequent pick: ECSW $r = 512$, $\kappa = 8$, BDF2 at $\Delta t = 1/60$ — a different scheme and step from ours, so basis, hyper-reduction and solver effects are not separated.
- **Caveats:** median error sometimes reverses (e.g. 3D W256: ours 0.57 % vs pick 1.11 %). Tuning budget: 2D unresolved against the factor-1.25 band (baseline/ours 1.16–1.73); 3D the baseline spent less (0.51–0.66).
- Coordinator cross-check: the lane's "ours" errors equal priority 1's independent export (R512 CN-R worst 0.02167 at 1024^2).





Consequence

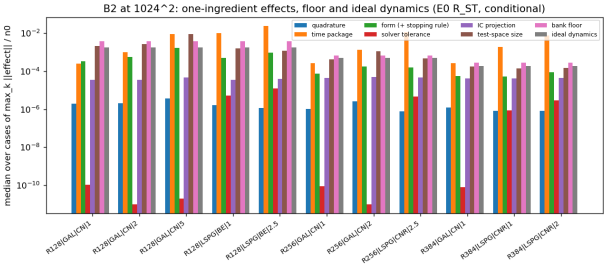
H-base ("matches or beats POD-based hyper-reduced ROMs at equal tuning budget") **fails** on this developmental evidence: a standard POD+ECSW/Q-DEIM model is faster and more accurate in the worst case on the box. With K2 also firing in both dimensions, the method's remaining distinctive

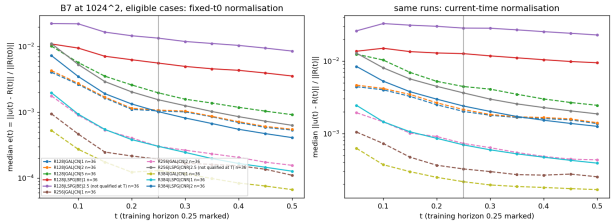
claims are mesh-independent cost and mesh-free evaluation; speed and accuracy advantages over standard alternatives are not supported for viscous Burgers on the box.

15 2026-10-19 — Priority 3: error budget, longer horizons, sampling adequacy

Lane `2026-10-18-jcp-budget`, last commit `d70608191`, mirror `9be95f4fb`. H200 jobs; 25.9 of 45 GPU-h; results audit 2 READY-FOR-REPORT, report audit 4 READY. New longer-horizon references (T = 0.5, 1.0) were generated with the E0 code in the lane's own namespace (dev/val cohorts only). Numbers quoted from the generated `report.md`; E0-conditional.

- **B2 error budget:** for the LSPG members the **time discretisation** is the dominant error at every mesh (256^2–4096^2); for GAL-CN at small r it is the bank floor. 7 of 10 members keep the same dominant ingredient at all three meshes. In 3D: floor for the W256 fixed-sweep members, time for the quasi-Newton ones. Coordinator cross-check: R384 CN-R at 256^2, time package 1.89×10^{-3} and floor 2.82×10^{-4}, identical to the combined lane's independent budget.
- **B7 longer horizons:** at $T = 0.5$ every 2D member is stable with no error growth (median W_2/W_1 0.17–0.67); some LSPG arms lose Q1 qualification. In 3D all four members are stable but fail Q2 at the longer horizon. At $T = 1.0$ (1024^2, 9 rankable cases) all members qualify and are stable; growth "none" is provisional (an unvalidated reference tier).
- **B8 sampling:** the 257^2 scoring lattice is adequate for the LSPG members relative to a 1025^2 route at every mesh; not for R384 GAL-CN. 3D sampling adequacy deferred.





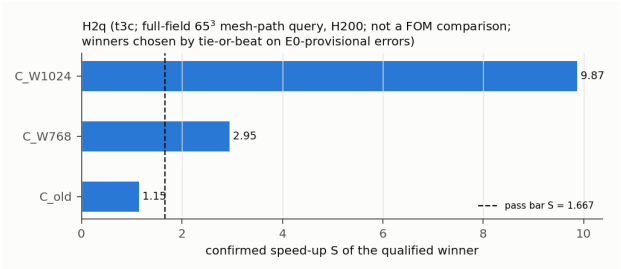
Corrections

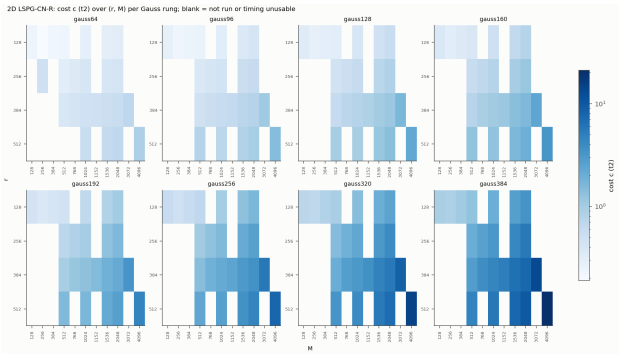
Retracted: Costs of E0's tight re-solve forced reruns without it (those cases non-rankable, A9). A later claim that disabling a tight retry cost rankable cases was itself corrected (A14.2: it cost none). A dominance tie rule compared only the top two candidates (A14.3; two counts changed, no label). A 3D reference mismatch traced to a last-bit inconsistency inside E0's store (A13).

16 2026-10-20 — Priority 3: test space, quadrature, rule families and a qualified 3D speed-up

Lane `2026-10-18-jcp-dims`, last commit `a99906fe6`, mirror `038cce675`. H200 jobs (`jax_backend=gpu`), 35.2 of 55 GPU-h, nothing deferred; results audit 2 ACCEPT, report audit 5 ACCEPT. Numbers quoted from the generated `report.md`; error-based verdicts are E0-conditional.

- **A4 (smallest qualifying quadrature, 2D):** 22 of 26 (r, M) cells reach a Gauss $m^\star$ (9,216–36,864 points); the 4 square cells ($M = r$) reach none.
- **A3 (2D factorial):** 128→256→384 improve, 384→512 does not — the same regression as priorities 1–2.
- **B6 (test space):** the deliberately weak $M = r$ control is unavailable — none of 108 square configurations passes solver qualification. A low mixed-frequency test set is stable but needs 4× more Gauss points; accuracy inconclusive.
- **B3 (rule families):** Fibonacci lattice $m^\star$ 4,181–28,657 in 2D; Gauss and lattice tie at $m^\star$ = 32,768 in 3D; Halton, Sobol and Monte Carlo are flagged by the cross-family gate; the bad-vector control fired in 2D and 3D.
- **2D cost elasticities:** about 0.65 in r, 0.53 in M, 0.65 in m.
- **A qualified 3D speed-up:** the 3D quadrature search found qualified configurations at every width (2–32 each). The cheapest that ties or beats each earlier comparator, M2_512 Gauss 32^3 QN-BDF2 at $\Delta t = 1/60$, is **2.95**× faster than the qualified W768 setup (pass), 9.87× faster than the unqualified W1024 setup, and only 1.15× faster than the old deployment (fails the 1.67× bar). Coordinator recheck from raw screen timings: 1.19×, 2.98×, 9.75×. This is ROM against ROM; it does not change K2.





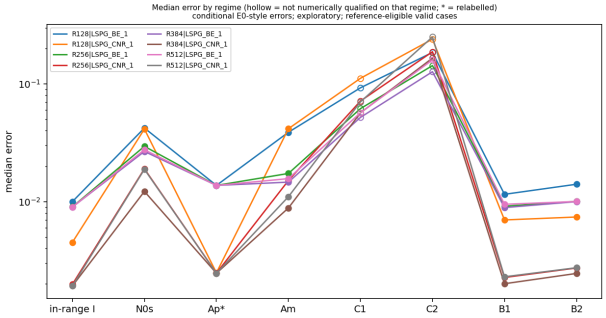
Corrections

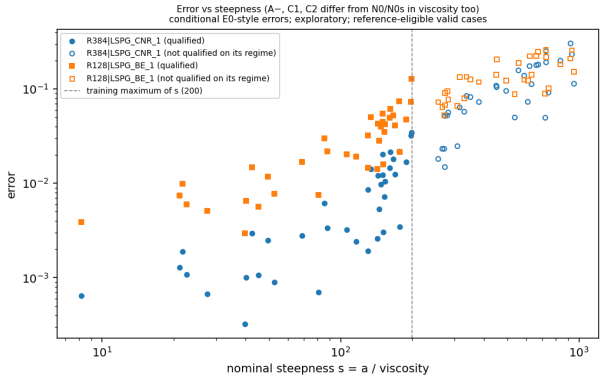
Retracted: A post-smoke relaxation of the certificate bar is now stated as such (A6.1/A9.1); the discrepancy figures first given for it were wrong and are replaced. Time estimates for two jobs were too low (reruns, A7, A10.2). The registered timing-identity checks had not been implemented before the first timing config; fixed before submission (A14). One job hit its time limit and its last setting was rerun (A16).

17 2026-10-20 — Priority 4: new regimes — where the frozen model stops working

Lane `2026-10-19-jcp-regimes`, last commit `70ce5f0f2`, mirror `ee52292fe`. 13 jobs (H200/A100), 14.9 of 60 GPU-h; results audit 2 CONFIRMED, report audit 4 READY. Seven freshly drawn 2D development regimes (seeds disjoint from every cohort; sealed/OOD untouched), new order-verified references, the 8 qualified LSPG configurations at 1024^2 (+1 GAL diagnostic). Exploratory development evidence; errors are E0-style conditional estimates. Numbers from the generated `report.md`.

- **Steeper fronts (C1, C2): not numerically qualified for any of the 8 configurations**; median errors 5.2–11 % (C1) and 12.7–25 % (C2), 6–131× the in-range baseline. The bank cannot represent them: the R384 representation floor rises from 0.03 % in range to 3.5 % (C1) and 9.9 % (C2). Error grows with steepness (Spearman $\rho = 0.93$).
- **Stiff corner of the training range (N0s):** all 8 degrade, 3–10×.
- **Viscosity below the range (A−):** CN-R degrades at R128 and R256; the rest inconclusive. Above the range (A+): holds or inconclusive, but confounded with steepness (its fronts are smoother).
- **New initial-condition families (B1, B2):** hold or inconclusive (ratios 0.99–1.64).
- Pseudo-spectral cross-check of the new references: 7/7 pass. Coordinator cross-check: the lane's in-range R128 BE worst error equals priority 1's (0.04843).





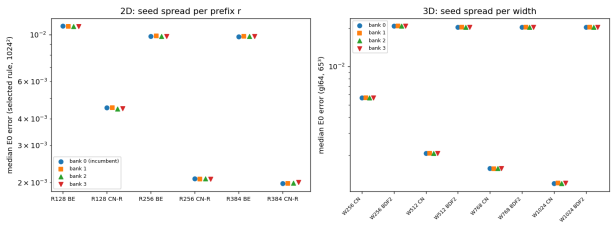
Corrections

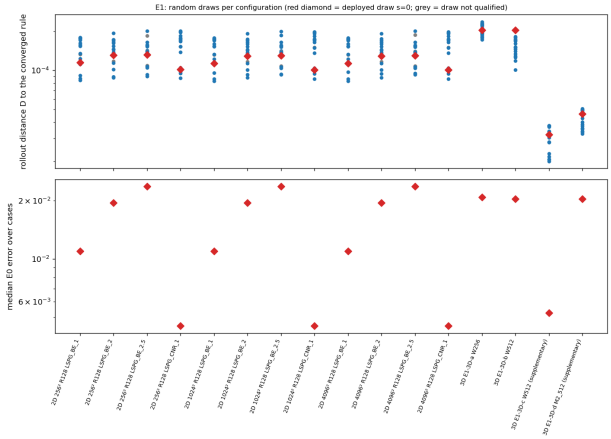
Retracted: The scoring first counted runs with solver budget exits as valid, contrary to the frozen rule; fixing it changed only the GAL diagnostic on C2 (now FAILS). An over-specified GPU-float fingerprint in the identity check (A7) and an exact-hash descriptor check across CPU types (A10) were relaxed to round-off before results were read. 3D regimes and the FOM comparison deferred.

18 2026-10-20 — Priority 4: robustness — random rules, retraining seeds, hardware

Lane `2026-10-19-jcp-robust`, last commit `cbf574373`, mirror `d49bd0085`. A100/H200 jobs (`jax_backend=gpu`), 46.4 of 95 GPU-h; every results audit accepted, report audit 3 ACCEPT. Numbers quoted from the generated `report.md`; errors E0-conditional.

- **E1 random quadrature draws (16 each):** in 3D both registered configurations are robust and insensitive (the deployed W512 draw is the least favourable of 16 but still qualifies). In 2D the R128 BE arms at $2\Delta t_0$ and $2.5\Delta t_0$ are "fragile" at all meshes — but every failing draw fails only on distance to its own successor rule; all stay within 2.02×10^{-4} of the independent target, and accuracy is insensitive in every cell.
- **E2 full retraining, seeds 1–3:** in 2D, 140 of 183 verdict rows are seed-robust, 21 seed-dependent, 22 inconclusive. **Mesh-independence of accuracy and the improvement with r survive retraining.** The selected quadrature rule changes with the seed, and the deployed R128 rule fails qualification on all three retrained banks (R384 on two) — rule selection is a per-bank step. In 3D (129 rows: 45/17/67), accuracy improves with width on every seed for the fixed Gauss rule; worst-case error at W1024 varies up to 1.9× across seeds (median 1.009×); W768–W1024 have almost no certified rules. Coordinator recount of the verdict files: identical counts.
- **E5 hardware (A100 vs H200, two jobs each):** 5 of 6 registered verdicts agree, 1 inconclusive, none contradicts; 4 of 10 cost ratios are hardware-dependent.





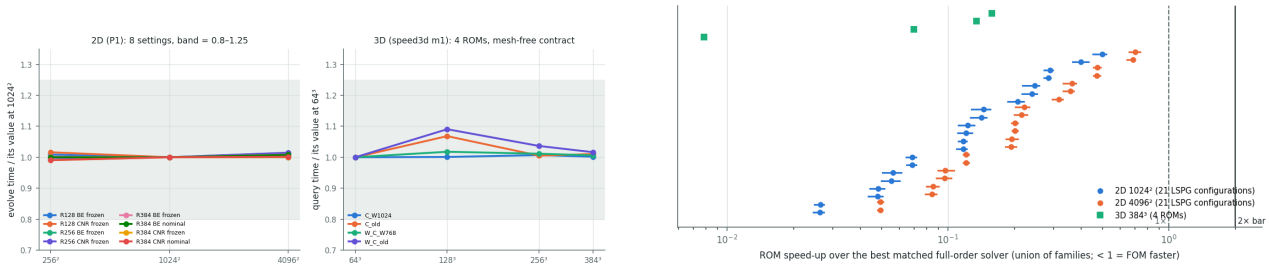
Corrections

Retracted: The first E2 2D roll-up used the wrong dominance rule; its counts (107/18/22) and the claim that R256 front membership is seed-dependent are withdrawn (A17). Report-generation errors (double-counted GPU-hours, a statement contradicting the verdict table, wrong glossary definitions, a misread data hash) were fixed (A19–A21). One job timed out and one was cancelled after under-estimated certificates.

19 2026-10-20 — Burgers 2D/3D campaign: final compilation

All ten lanes of the campaign (`reports/2026-10-17-full-experiment-plan/`) are finished and compiled by `final/make_final.py` into `final/report.md` (every number traced in `PROVENANCE.json`; Codex final audit round 12 CLEAN). Total compute 168.8 GPU-h. Development/validation cohorts only; sealed and reserved-OOD cohorts unopened; E0-conditional errors.

- **The advisor's question.** Solve cost does not depend on the mesh (2D and 3D, both readings of k, across retraining seeds and GPU types); accuracy does not depend on the mesh; accuracy improves with bank width up to $r = 384$ (not to 512); kernel cost does not follow the registered operation-count exponents.
- **Competition.** K2 fires in 2D and 3D: a coarse spectral solver matches the ROM's accuracy faster. H-base fails: a tuned POD + ECSW/Q-DEIM model dominates our qualified 2D configurations at 256^2 and 1024^2.
- **Robustness and generality.** Not qualified on steeper fronts than in training (the bank cannot represent them); degrades in the stiff corner of its own range. Mesh-independence and the r dial survive retraining; the quadrature rule must be selected per bank.
- **What the paper can claim:** mesh-independent cost, mesh-free (continuum) evaluation and memory; speed and accuracy advantages over standard alternatives are not supported for viscous Burgers on the box.



Branches of all ten lanes await the user's merge decision.

20 2026-10-22 — Improvement campaign, wave 1

Coordinator campaign `reports/2026-10-21-improvement-campaign/` (lanes `exp/2026-10-21-imp-*`). Errors are conditional E0 estimates; POD comparisons are developmental and in-sample. Wave 1 used about 8.3 GPU-h.

- **E5, faster step: stop rule fires.** Precontracted Galerkin with ETDRK4 evolves 14–24× faster than LSPG CN-R at $r = 384$ and wins 36 of 38 cases, but its worst error is slightly higher (1.82 % vs 1.78 %). POD + ECSW with the same step is still faster at $r = 384$.
- **E6, cheaper nonlinear term: both stop rules fire.** Smaller rules either fail the quadrature check or lose accuracy; the best gain is 1.24×.
- **E7, multi-mesh cost: kill criterion fires.** A continuous POD built once and resampled matches our worst error at about 0.1× our full cost on every workload; the comparative multi-mesh cost claim is dropped.
- **E3, variable coefficient: gate fails.** The fastest tuned FOM (exponential integrator with a constant-coefficient split) is not slower than our box ROM (0.86–0.92×); no ROM trained.
- **E8 Phase A (CPU):** the decoded bank fields carry a large wall-concentrated sine tail; a bank+sine hybrid does not reach the steep-front floor bar. Phase B (coverage retrain) is running; E2 (geometry) and E1 (box family width) are running.

